

FAULT CODE 227 - Sensor Supply 2 Circuit - Voltage Above Normal or Shorted to High Source

Warnings and Cautions

CAUTION

To reduce the possibility of damaging a new ECM, all other active fault codes must be investigated prior to replacing the ECM.

Troubleshooting Summary

STEPS		SPECIFICATIONS
STEP 1.	Check the fault codes.	
	STEP 1A. Check for an active fault code.	Fault Code 227 active?
STEP 2.	Check the ECM, OEM harness, and engine harness.	
	STEP 2A.. Inspect the ECM, OEM harness, and engine harness connector pins.	Dirty or damaged pins?
	STEP 2B. Check for a pin-to-pin short circuit in the OEM harness or engine harness.	Greater than 100k ohms?
	STEP 2C. Check for a short circuit in the unswitched battery supply power harness.	Greater than 100k ohms?
STEP 3.	Clear the fault codes.	
	STEP 3A. Disable the fault code.	Fault Code 227 inactive?
	STEP 3B. Clear the inactive fault codes.	All fault codes cleared?

STEP 1. Check the fault codes.

STEP 1A. Check for an active fault code.

Conditions :

- Turn keyswitch ON.
- Connect INSITE™ electronic service tool.

Action	Specification/Repair	Next Step
Check for an active fault code. • Use INSITE™ electronic service tool to read the fault codes.	Fault Code 227 active? YES	2A
	Fault Code 227 active? NO	Use the following procedure for an inactive or intermittent fault code. Refer to Procedure 019-362 in Section 19. (/qs3/pubsys2/xml/en/procedures/99/99-019-362.html)

STEP 2. Check the ECM, OEM harness, and engine harness.

STEP 2A. Inspect the ECM, OEM harness, and engine harness connector pins.

Conditions :

- Turn keyswitch OFF.
- Disconnect the accelerator pedal or lever position sensor connector from the OEM harness connector, if equipped.
- Disconnect the speed bias connector from the OEM harness connector, if equipped.
- Disconnect the gain adjust potentiometer connector from the OEM harness connector, if equipped.
- Disconnect the engine harness connector from the ECM 60-pin OEM port connector.
- Disconnect the engine harness connector from the 31-pin OEM connector.

Action	Specification/Repair	Next Step
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Inspect the engine harness and sensor connector pins for the following: • Loose connector • Corroded pins • Bent or broken pins • Pushed back or expanded pins • Moisture in or on the connector • Missing or damaged connector seals • Dirt or debris in or on the connector pins • Connector shell broken • Wire insulation damage • Damaged connector locking tab. Use the following procedure for general inspection techniques. Refer to Procedure 019-361 in Section 19. (/qs3/pubsys2/xml/en/procedures/99/99-019-361.html)	Dirty or damaged pins? YES Repair : Clean the connector and pins. Repair the damaged harness, connector, or pins if possible. Refer to the circuit diagram or wiring diagram for all engine harness interconnections. • Refer to Procedure 019-071 in Section 19. (/qs3/pubsys2/xml/en/procedures/99/99-019-071.html) • Refer to Procedure 019-204 in Section 19. (/qs3/pubsys2/xml/en/procedures/99/99-019-204.html) • Refer to Procedure 019-043 in Section 19. (/qs3/pubsys2/xml/en/procedures/123/123-019-043.html) • Refer to the OEM service manual for accelerator pedal or lever position sensor repair instructions. Replace the ECM. Refer to Procedure 019-031 in Section 19. (/qs3/pubsys2/xml/en/procedures/123/123-019-031.html)	3A
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Dirty or damaged pins?

NO

2B

STEP 2B. Check for a pin-to-pin short circuit in the OEM harness or engine harness.

Conditions :

- Turn keyswitch OFF.
- Disconnect the engine harness connector from the ECM 60-pin OEM port connector.
- Disconnect the accelerator pedal or lever position sensor connector from the OEM harness connector, if equipped.
- Disconnect the speed bias connector from the OEM harness connector, if equipped.
- Disconnect the gain adjust potentiometer connector from the OEM harness connector, if equipped.

Action

Specification/Repair

Next Step

	Greater than 100k ohms? YES	2C
	Greater than 100k ohms? NO Repair : A short circuit has been detected in the 5 volt SUPPLY (sensor supply 2) wire. Troubleshoot all harnesses connected in series to determine which contains the pin-to-pin short. Refer to the circuit diagram or wiring diagram for all harness interconnections. Replace the damaged section of harness. • Refer to Procedure 019-071 in Section 19. (/qs3/pubsys2/xml/en/procedures/99/99-019-071.html) • Refer to Procedure 019-199 in Section 19. (/qs3/pubsys2/xml/en/procedures/99/99-019-199.html) • Refer to Procedure 019-043 in Section 19. (/qs3/pubsys2/xml/en/procedures/123/123-019-043.html)	3A

STEP 2C. Check for a short circuit in the unswitched battery supply power harness.

Conditions :

- Turn keyswitch OFF.
- Disconnect the engine harness connector from the ECM 60-pin OEM port connector.
- Disconnect the accelerator pedal or lever position sensor connector from the OEM harness connector, if equipped.
- Disconnect the speed bias connector from the OEM harness connector, if equipped.
- Disconnect the gain adjust potentiometer connector from the OEM harness connector, if equipped.
- Disconnect the engine power harness connector from the ECM connector.

Action	Specification/Repair	Next Step
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Check for a short circuit to battery. Measure the resistance between the 5 volt SUPPLY (sensor supply 2) pin in the engine harness ECM connector and the ECM battery SUPPLY (+) pin or the power harness ECM connector. Refer to the circuit diagram or wiring diagram for connector pin identification. Use the following procedure for general resistance measurement techniques. Refer to Procedure 019-360 in Section 19. (/qs3/pubsys2/xml/en/procedures/99/99-019-360.html)	Greater than 100k ohms? YES Repair : Replace the ECM. Refer to Procedure 019-031 in Section 19. (/qs3/pubsys2/xml/en/procedures/123/123-019-031.html)	3A
	Greater than 100k ohms? NO Repair : A short circuit to the battery has been detected in the OEM harness or engine harness. - Repair the OEM harness. Refer to Procedure 019-071 in Section 19. (/qs3/pubsys2/xml/en/procedures/99/99-019-071.html) - Repair the engine harness. Refer to Procedure 019-043 in Section 19. (/qs3/pubsys2/xml/en/procedures/123/123-019-043.html)	3A

STEP 3. Clear the fault codes.

STEP 3A. Disable the fault code.

Conditions :

- Connect all components.
- Turn keyswitch ON.
- Connect INSITE™ electronic service tool.

Action	Specification/Repair	Next Step
Disable the fault code. • Use INSITE™ electronic service tool to verify the fault code is inactive.	Fault Code 227 inactive? YES	3B
	Fault Code 227 inactive? NO Repair : Return to the troubleshooting steps or contact a Cummins® Authorized Repair Location if all steps have been completed and checked again.	1A

STEP 3B. Clear the inactive fault codes.**Conditions :**

- Connect all components.
- Turn keyswitch ON.
- Connect INSITE™ electronic service tool.

Action	Specification/Repair	Next Step
Clear the inactive fault codes. • Use INSITE™ electronic service tool to clear the inactive fault codes.	All fault codes cleared? YES	Repair complete.
	All fault codes cleared? NO Repair : Troubleshoot any remaining active fault codes.	Go to the appropriate troubleshooting steps.

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